

--- FILE: Cinema.php ---

```
<?php
declare(strict_types=1);
class Cinema {
    // Properties
    private array $movies = [];
    // Constructor
    function __construct (
        private string $name,
        private string $city
    ){
    }
    // Getters
    public function getName(): string {
        return $this->name;
    }
    public function getCity(): string {
        return $this->city;
    }
    public function getMovies(): array {
        return $this->movies;
    }
    public function getProperty(string $property): string {
        return match ($property) {
            "name" => $this->name,
            "city" => $this->city,
            default => throw new InvalidArgumentException("$property is not a
valid argument")
        };
    }
    // Methods
    public function addMovie(int $id, Movie $m): void {
        $this->movies[$id] = $m;
    }
}
```

--- FILE: CinemaChain.php ---

```
<?php
require_once('Movie.php');
require_once('Cinema.php');
class CinemaChain {
    // Properties
    private array $movies = [];
    private array $cinemas = [];
    // Constructor
    function __construct(array $parMovies, array $parCinemas, array
$parRelationships) {
        if (count($parMovies) == 0) throw new InvalidArgumentException
('$parMovies is empty');
        if (count($parCinemas) == 0) throw new InvalidArgumentException
('$parCinemas is empty');
        if (count($parRelationships) == 0) throw new InvalidArgumentException
('$parRelationships is empty');
```

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        foreach($parMovies as $m) {
            $this->movies[$m["id"]] = new Movie(
                $m["title"],
                $m["duration"],
                $m["director"]
            );
        };
        foreach($parCinemas as $c) {
            $this->cinemas[$c["id"]] = new Cinema(
                $c["name"],
                $c["city"]
            );
        };
        foreach($parRelationships as $r) {
            $cinema = $this->cinemas[$r["c_id"]];
            $movie = $this->movies[$r["m_id"]];
            $cinema->addMovie($r["m_id"], $movie);
            $movie->addCinema($r["c_id"], $cinema);
        }
    }
    // Getters
    public function getMovies(): array {
        return $this->movies;
    }
    public function getCinemas(): array {
        return $this->cinemas;
    }
    // Methods
    public function getMoviesBy(string $property, string|int $value): array {
        $movies = [];
        foreach($this->movies as $m) {
            if ($m->getProperty($property) === $value) array_push($movies, $m);
        }
        return $movies;
    }
    public function getLongestMovie(string $cinemaName): ?Movie {
        foreach ($this->cinemas as $c) {
            if ($c->getName() === $cinemaName) {
                $movies = $c->getMovies();
                if (count($movies) === 0) return null;
                $longestMovie;
                $longestDuration = -1;
                foreach ($movies as $m) {
                    if ($m->getDuration() > $longestDuration) {
                        $longestMovie = $m;
                        $longestDuration = $longestMovie->getDuration();
                    }
                }
                return $longestMovie;
            }
        }
        return null;
    }
}

```

--- FILE: Movie.php ---

```
<?php
declare(strict_types=1);
class Movie {
    // WARNING: circular references.
    // Use a loop for display; avoid using print_r() to prevent recursion
    issues.
    // Properties
    private array $cinemas = [];
    // Constructor
    function __construct(
        private string $title,
        private int $duration,
        private string $director
    ){
    // Getters
    public function getTitle(): string {
        return $this->title;
    }
    public function getDuration(): int {
        return $this->duration;
    }
    public function getDirector(): string {
        return $this->director;
    }
    public function getCinemas(): array {
        return $this->cinemas;
    }
    public function getProperty(string $property): int|string {
        return match ($property) {
            "title" => $this->title,
            "duration" => $this->duration,
            "director" => $this->director,
            default => throw new InvalidArgumentException("$property is not a
valid argument")
        };
    }
    // Methods
    public function addCinema(int $id, Cinema $c): void {
        $this->cinemas[$id] = $c;
    }
}
```

--- FILE: cinesa_data.php ---

```
<?php
/*
SELECT movies.id, movies.title, movies.duration, movies.director
FROM movies
ORDER BY movies.id ASC;
*/
$movies_ordered_by_id = [
```

```

    ["id" => 1, "title" => "Super Mario Galaxy: La película", "duration" => 98,
"director" => "Aaron Horvath"],
    ["id" => 2, "title" => "Proyecto Salvación", "duration" => 156, "director"
=> "Phil Lord"],
    ["id" => 3, "title" => "La momia de Lee Cronin", "duration" => 134,
"director" => "Lee Cronin"],
    ["id" => 4, "title" => "Amarga Navidad", "duration" => 111, "director" =>
"Pedro Almodóvar"],
    ["id" => 5, "title" => "Iron Lung: Océano de sangre", "duration" => 86,
"director" => "Mark Fischbach"],
    ["id" => 6, "title" => "Mortal Kombat II", "duration" => 116, "director" =>
"Simon McQuoid"],
    ["id" => 7, "title" => "El diablo viste de Prada 2", "duration" => 120,
"director" => "David Frankel"],
    ["id" => 8, "title" => "Hanna i els Nadals oblidats", "duration" => 88,
"director" => "Elena Ruiz"],
    ["id" => 9, "title" => "La familia Benetón +2", "duration" => 87, "director"
=> "Joaquín Mazón"],
    ["id" => 10, "title" => "Kill Bill: The Whole Bloody Affair", "duration" =>
275, "director" => "Quentin Tarantino"],
    ["id" => 11, "title" => "ONE OK ROCK DETOX JAPAN TOUR", "duration" => 118,
"director" => "Naoto Amazutsumi"],
    ["id" => 12, "title" => "Boulevard", "duration" => 113, "director" => "Sonia
Méndez"],
    ["id" => 13, "title" => "Noche de bodas 2", "duration" => 108, "director" =>
"Matt Bettinelli-Olpin"],
    ["id" => 14, "title" => "Shelter: El protector", "duration" => 107,
"director" => "Alexis Morante"],
    ["id" => 15, "title" => "La mujer más rica del mundo", "duration" => 122,
"director" => "Thierry Klifa"],
    ["id" => 16, "title" => "El chico de los pantalones rosas", "duration" =>
115, "director" => "Margherita Ferri"],
    ["id" => 17, "title" => "A una isla de ti", "duration" => 102, "director" =>
"Simón Mesa Soto"],
    ["id" => 18, "title" => "Torrente Presidente", "duration" => 102, "director"
=> "Santiago Segura"],
    ["id" => 19, "title" => "Hoppers", "duration" => 104, "director" => "Daniel
Chong"],
    ["id" => 20, "title" => "BTS WORLD TOUR 'ARIRANG'", "duration" => 115,
"director" => "Simon Hilton"],
    ["id" => 21, "title" => "Five Nights at Freddy's 2", "duration" => 104,
"director" => "Emma Tammi"],
    ["id" => 22, "title" => "Golpes", "duration" => 102, "director" => "Rafael
Cobos"],
    ["id" => 23, "title" => "Dune: Parte tres", "duration" => 165, "director" =>
"Denis Villeneuve"],
    ["id" => 24, "title" => "Me has robado el corazón", "duration" => 98,
"director" => "Chus Gutiérrez"],
    ["id" => 25, "title" => "Michael", "duration" => 150, "director" => "Antoine
Fuqua"]
];
/*
SELECT cinemas.id, cinemas.name, cinemas.city
FROM cinemas

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ORDER BY cinemas.id ASC;
*/
$cinemas_ordered_by_id = [
    ["id" => 1, "name" => "Cinesa Diagonal", "city" => "Barcelona"],
    ["id" => 2, "name" => "Cinesa Diagonal Mar", "city" => "Barcelona"],
    ["id" => 3, "name" => "Cinesa SOM Multiespai", "city" => "Barcelona"],
    ["id" => 4, "name" => "Cinesa LUXE La Farga", "city" => "L'Hospitalet"],
    ["id" => 5, "name" => "Cinesa Parc Vallès", "city" => "Terrassa"],
    ["id" => 6, "name" => "Cinesa LUXE Barnasud", "city" => "Gavà"]
];
/*
SELECT cinemas.id AS c_id, movies.id AS m_id
FROM cinemas
JOIN movies ON cinemas.id = movies.cinema_id
WHERE cinemas.region = 'Catalonia'
ORDER BY cinemas.id ASC;
*/
$cinema_movie_mapping = [
    ["c_id" => 1, "m_id" => 1],
    ["c_id" => 1, "m_id" => 2],
    ["c_id" => 1, "m_id" => 3],
    ["c_id" => 1, "m_id" => 4],
    ["c_id" => 1, "m_id" => 5],
    ["c_id" => 1, "m_id" => 6],
    ["c_id" => 1, "m_id" => 7],
    ["c_id" => 1, "m_id" => 8],
    ["c_id" => 2, "m_id" => 1],
    ["c_id" => 2, "m_id" => 9],
    ["c_id" => 2, "m_id" => 10],
    ["c_id" => 2, "m_id" => 11],
    ["c_id" => 2, "m_id" => 12],
    ["c_id" => 2, "m_id" => 3],
    ["c_id" => 2, "m_id" => 13],
    ["c_id" => 2, "m_id" => 14],
    ["c_id" => 3, "m_id" => 1],
    ["c_id" => 3, "m_id" => 15],
    ["c_id" => 3, "m_id" => 16],
    ["c_id" => 3, "m_id" => 17],
    ["c_id" => 3, "m_id" => 18],
    ["c_id" => 3, "m_id" => 4],
    ["c_id" => 3, "m_id" => 2],
    ["c_id" => 3, "m_id" => 19],
    ["c_id" => 4, "m_id" => 1],
    ["c_id" => 4, "m_id" => 3],
    ["c_id" => 4, "m_id" => 9],
    ["c_id" => 4, "m_id" => 20],
    ["c_id" => 4, "m_id" => 2],
    ["c_id" => 4, "m_id" => 18],
    ["c_id" => 4, "m_id" => 19],
    ["c_id" => 4, "m_id" => 5],
    ["c_id" => 5, "m_id" => 1],
    ["c_id" => 5, "m_id" => 10],
    ["c_id" => 5, "m_id" => 21],
    ["c_id" => 5, "m_id" => 22],

```

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        ["c_id" => 5, "m_id" => 2],
        ["c_id" => 5, "m_id" => 3],
        ["c_id" => 5, "m_id" => 18],
        ["c_id" => 5, "m_id" => 23],
        ["c_id" => 6, "m_id" => 24],
        ["c_id" => 6, "m_id" => 1],
        ["c_id" => 6, "m_id" => 9],
        ["c_id" => 6, "m_id" => 4],
        ["c_id" => 6, "m_id" => 19],
        ["c_id" => 6, "m_id" => 18],
        ["c_id" => 6, "m_id" => 2],
        ["c_id" => 6, "m_id" => 25]
];

```

--- FILE: ejercici_1.php ---

```

<?php
declare(strict_types=1);
require_once('cinesa_data.php');
require_once('CinemaChain.php');
$cinesa = new CinemaChain(
    $movies_ordered_by_id,
    $cinemas_ordered_by_id,
    $cinema_movie_mapping
);
foreach ($cinesa->getCinemas() as $c) {
echo '-----' . PHP_EOL;
    $longestMovie = $cinesa->getLongestMovie($c->getName());
    printf("%s\n", $c->getName());
    printf("%s\n", $c->getCity());
    printf("%-50s%-30s%10s\n", 'TITLE', 'DIRECTOR', 'DURATION');
    foreach ($c->getMovies() as $m) {
        $title = $m->getTitle();
        $director = $m->getDirector();
        echo $title . str_repeat(' ', 50 - mb_strlen($title));
        echo $director . str_repeat(' ', 30 - mb_strlen($director));
        if ($m === $longestMovie) {
            printf("%3s%7d\n", '-->', $m->getDuration());
        } else {
            printf("%10d\n", $m->getDuration());
        }
    }
}
echo '-----' . PHP_EOL;
foreach ($cinesa->getMoviesBy("director", "Pedro Almodóvar") as $m) {
    echo $m->getTitle() . PHP_EOL;
};
echo '-----' . PHP_EOL;
foreach ($cinesa->getMoviesBy("duration", 102) as $m) {
    echo $m->getTitle() . PHP_EOL;
};echo '-----' . PHP_EOL;

```

--- FILE: diagram.puml ---

```

@startuml
skinparam classAttributeIconSize 0
class Movie {
    - title: string
    - duration: int
    - director: string
    - cinemas: array
    + getTitle(): string
    + getDuration(): int
    + getDirector(): string
    + getCinemas(): array
    + getProperty(property: string): int|string
    + addCinema(id: int, c: Cinema): void
}
class Cinema {
    - name: string
    - city: string
    - movies: array
    + getName(): string
    + getCity(): string
    + getMovies(): array
    + getProperty(property: string): string
    + addMovie(id: int, m: Movie): void
}
class CinemaChain {
    - movies: array
    - cinemas: array
    + getMovies: array
    + getCinemas: array
    + getMoviesBy(property: string, value: string|int): array
    + getLongestMovie(cinemaName: string): ?Movie
}
CinemaChain "1" *-- "1..*" Movie
CinemaChain "1" *-- "1..*" Cinema
Movie "1..*" o--o "1..*" Cinema
@enduml

```